AQUABC Model

Parameter & Constant Reference

Based on WCONST_04.txt (latest configuration)

Total constants: 323

Generated: February 2026

This document provides a complete reference for all model constants in the AQUABC pelagic ecological model (file WCONST_04.txt). Constants are organised by functional category. Each section shows the constant number, name, current value, measurement unit, and description. A value-selection rationale explains *why* specific values were chosen, citing literature ranges. The AQUABC model simulates phytoplankton dynamics (diatoms, cyanobacteria, other algae, Nostocales), zooplankton grazing, organic matter cycling, nutrient dynamics (N, P, Si), dissolved oxygen, redox chemistry (Fe, Mn), methane, hydrogen sulphide, and pH/alkalinity. Temperature effects use the Cardinal Temperature Model with Inflection (CTMI; Rosso et al. 1993). Nutrient limitation follows Monod (Michaelis-Menten) kinetics. The multi-electron-acceptor mineralisation scheme follows Van Cappellen & Wang (1996).

File format (Fortran free-format read): <integer_index> <constant_name> <real_value> ! <description>

Table of Contents

General / Physical Parameters

General physical and optical parameters controlling aeration and light attenuation through the water column.

Value Selection Rationale:

Aeration rate K_A is set to -1.0 (auto-calculated from wind and depth using O'Connor-Dobbins or similar). THETA_K_A = 1.04 follows the standard Arrhenius range 1.02-1.06 (Chapra, 1997). XKC = 0.08 m^-1/(ug Chla/L) is at the upper end of the 0.01-0.08 range reported by Kirk (1994), suitable for eutrophic systems. PHIMX = 720 mg C/mol photon is within the literature range 700-1000 (Falkowski & Raven, 2007).

#	Constant Name	Value	Unit	Description
1	`K_A`	-1.0	*1/day*	Aeration coefficient (if negative calculates internally)
2	`THETA_K_A`	1.04	-	Temperature correction factor for aeration
3	`XKC`	0.08	*1/m/(ug Chla/L)*	fixme. Light extinction per chlorophyl unit,(mcg Chla/l/m) for calculation of saturation for Smith. Should be different for different phyto?
4	`PHIMX`	720.00	*mg C/mol photon*	Quantum yield const. mg C/mole photon

Literature References:

[1] Chapra, S.C. (1997)
Surface Water-Quality Modeling. McGraw-Hill, New York.
Relevance: Aeration coefficients and temperature correction theta = 1.024 for reaeration.

[2] Bowie, G.L., Mills, W.B., Porcella, D.B. et al. (1985)
Rates, Constants, and Kinetics Formulations in Surface Water Quality Modeling (2nd ed.). EPA/600/3-85/040. U.S. EPA, Athens, GA.
Relevance: Comprehensive compilation of rate constants for surface water quality models.

[3] Kirk, J.T.O. (1994)
Light and Photosynthesis in Aquatic Ecosystems, 2nd ed. Cambridge Univ. Press.
Relevance: Light extinction coefficient per unit chlorophyll (XKC ~ 0.01-0.08 m^-1/(ug Chla/L)).

[4] Falkowski, P.G. & Raven, J.A. (2007)
Aquatic Photosynthesis, 2nd ed. Princeton Univ. Press.
Relevance: Quantum yield constant PHIMX (700-1000 mg C/mole photon).

Diatoms (Bacillariophyceae)

Growth, respiration, mortality, nutrient limitation, hypoxia stress, and stoichiometry for diatoms -- siliceous phytoplankton that dominate spring blooms in temperate lakes and estuaries.

Value Selection Rationale:

KG_DIA_OPT_TEMP = 3.7 /day: within 2-4 /day at optimal T (Eppley, 1972; Reynolds, 2006). CTMI cardinal temperatures: OPT_TEMP_LR = T_min = 1 degC, OPT_TEMP_UR = T_opt = 24 degC, KAPPA_OVER_OPT_TEMP = T_max = 35 degC (Bernard & Rémond, 2012; Rosso et al. 1993). Note: KAPPA_UNDER_OPT_TEMP is passed to GROWTH_AT_TEMP but unused in the CTMI equation (retained for backward compatibility). EFF_DIA_GROWTH = 0.95 means only 5% of gross growth lost to maintenance. KR_DIA_20 = 0.05 /day and KD_DIA_20 = 0.12 /day within ranges 0.03-0.10 and 0.05-0.20 (Reynolds, 2006). Hypoxia stress: FAC_HYPOX = THETA^(EXPON*(DO_threshold - DO)); EXPON has units L/mg O2 so the exponent is dimensionless. Half-saturations KHS_DIN=0.01, KHS_DIP=0.005, KHS_DSi=0.013 mg/L are typical for nutrient-rich systems (Jorgensen et al., 1991). N:C=0.22, P:C=0.024 follow Redfield (1958) with slight N enrichment; Si:C=0.25 follows Brzezinski (1985). C:Chla=30 is in the range 20-100 (Geider et al., 1997).

#	Constant Name	Value	Unit	Description
5	KG_DIA_OPT_TEMP	3.7	*1/day*	3.1 Diatoms Growth rate
6	DIA_OPT_TEMP_LR	1.0	*deg C*	Diatoms CTMI T_min (minimum cardinal temperature)
7	DIA_OPT_TEMP_UR	24.0	*deg C*	Diatoms CTMI T_opt (optimal temperature)
8	EFF_DIA_GROWTH	0.95	-	Diatoms Effective growth. (1-EG)*growth - losses for respiration and excretion
9	KAPPA_DIA_UNDER_OPT_TEMP	0.0	*deg C*	(unused, kept for backward compat)
10	KAPPA_DIA_OVER_OPT_TEMP	35.0	*deg C*	Diatoms CTMI T_max (maximum cardinal temperature)
11	KR_DIA_20	0.05	*1/day*	Diatoms Respiration rate
12	THETA_KR_DIA	1.04	-	Diatoms Temperature correction for basal respiration rate
13	KD_DIA_20	0.12	*1/day*	Diatoms Mortality rate
14	THETA_KD_DIA	1.02	-	Diatoms Temperature correction for Mortality rate
15	KHS_DIN_DIA	0.010	*mg N/L*	Diatoms Half saturation growth for DIN
16	KHS_DIP_DIA	0.005	*mg P/L*	Diatoms Half saturation growth for DIP
17	KHS_DSi_DIA	0.013	*mg Si/L*	Diatoms Half saturation growth for DSi
18	KHS_O2_DIA	0.60	*mg O2/L*	Diatoms Half saturation growth for O2
19	FRAC_DIA_EXCR	0.30	-	Diatoms Fraction of excretion in metabolism rate
20	I_S_DIA	100.00	*langleys*	Diatoms Light saturation (langleys)
21	DO_STR_HYPOX_DIA_D	0.70	*mg O2/L*	Diatoms Dissolved oxygen stress in oxygen units (mortality increase below this value exponentialy
22	THETA_HYPOX_DIA_D	1.20	-	Diatoms Multiplier of the exponent for Dissolved oxygen stress
23	EXPON_HYPOX_DIA_D	1.04	*L/mg O2*	Diatoms Exponent constant for Dissolved oxygen stress
24	DIA_N_TO_C	0.22	*mg N/mg C*	Diatoms Nitrogen to Carbon ratio
25	DIA_P_TO_C	0.024	*mg P/mg C*	Diatoms Phosphorus to Carbon ratio
26	DIA_Si_TO_C	0.25	-	Diatoms Silica to Carbon ratio
27	DIA_O2_TO_C	2.66	*mg O2/mg C*	Diatoms Oxygen to Carbon ratio for respiration
28	DIA_C_TO_CHLA	30.00	*mg C/mg Chla*	Diatoms Carbon to Chlorophil a ratio

Literature References:

[1] Eppley, R.W. (1972)
Temperature and phytoplankton growth in the sea. Fishery Bulletin, 70(4), 1063-1085.
Relevance: Maximum growth rate as a function of temperature; diatom mu_max ~ 2-4 /day at optimal T.

[2] Rosso, L., Lobry, J.R., & Flandrois, J.P. (1993)
An unexpected correlation between cardinal temperatures of microbial growth highlighted by a new model. J. Theoretical Biology, 162(4), 447-463.
Relevance: Cardinal Temperature Model with Inflection (CTMI): T_min, T_opt, T_max framework used for all phytoplankton temperature responses in AQUABC.

[3] Bernard, O. & Rémond, B. (2012)
Validation of a simple model accounting for light and temperature effect on microalgal growth. Bioresource Technology, 123, 520-527.
Relevance: Validated CTMI with three cardinal temperatures for various microalgae.

[4] Brzezinski, M.A. (1985)
The Si:C:N ratio of marine diatoms: interspecific variability and the effect of some environmental variables. J. Phycology, 21(3), 347-357.
Relevance: Si:C ratio for diatoms ~ 0.13-0.47 (model uses 0.25); Si:N:P = 15:16:1.

[5] Redfield, A.C. (1958)
The biological control of chemical factors in the environment. American Scientist, 46(3), 205-221.
Relevance: Redfield ratio C:N:P = 106:16:1 (N:C ~ 0.17, P:C ~ 0.024 by mass). Model N:C = 0.22 and P:C = 0.024 are within reported ranges.

[6] Geider, R.J., MacIntyre, H.L., & Kana, T.M. (1997)
Dynamic model of phytoplankton growth and acclimation. Marine Ecology Progress Series, 148, 187-200.
Relevance: C:Chl-a ratio 20-100 mg C / mg Chla (model uses 30 for diatoms).

[7] Reynolds, C.S. (2006)
The Ecology of Phytoplankton. Cambridge Univ. Press.
Relevance: Diatom respiration 0.03-0.10 /day; mortality 0.05-0.20 /day; KHS_DIN 0.005-0.05 mg/L.

Non-Fixing Cyanobacteria

Parameters for non-nitrogen-fixing cyanobacteria (e.g. Microcystis, Planktothrix). Warm-water specialists with higher optimal temperatures.

Value Selection Rationale:

KG_CYN_OPT_TEMP = 2.4 /day: within 1.0-3.5 /day (Reynolds, 2006). CTMI parameters: OPT_TEMP_LR = T_min = 15, OPT_TEMP_UR = T_opt = 26, KAPPA_OVER_OPT_TEMP = T_max = 38 degC (Robarts & Zohary, 1987; Rosso et al. 1993). KAPPA_UNDER_OPT_TEMP is unused in the CTMI equation. KHS_DIN=0.009, KHS_DIP=0.008 mg/L are low, reflecting cyanobacterial competitive advantage at low nutrients (Jorgensen et al., 1991). C:Chla=40 reflects higher pigment packaging than diatoms.

#	Constant Name	Value	Unit	Description
29	`KG_CYN_OPT_TEMP`	2.4	*1/day*	Non-fixing cyanobacteria Growth rate
30	`CYN_OPT_TEMP_LR`	15.0	*deg C*	Non-fixing cyanobacteria CTMI T_min
31	`CYN_OPT_TEMP_UR`	26.0	*deg C*	Non-fixing cyanobacteria CTMI T_opt
32	`EFF_CYN_GROWTH`	0.95	-	Non-fixing cyanobacteria Effective growth. (1-EG)*growth - losses for respiration and excretion
33	`KAPPA_CYN_UNDER_OPT_TEMP`	0.0	*deg C*	(unused)
34	`KAPPA_CYN_OVER_OPT_TEMP`	38.0	*deg C*	Non-fixing cyanobacteria CTMI T_max
35	`KR_CYN_20`	0.06	*1/day*	Non-fixing cyanobacteria Respiration rate
36	`THETA_KR_CYN`	1.04	-	Non-fixing cyanobacteria Temperature correction for respiration rate
37	`KD_CYN_20`	0.125	*1/day*	Non-fixing cyanobacteria Mortality rate
38	`THETA_KD_CYN`	1.05	-	Non-fixing cyanobacteria Temperature correction for Mortality rate
39	`KHS_DIN_CYN`	0.009	*mg N/L*	Non-fixing cyanobacteria Half saturation growth for DIN
40	`KHS_DIP_CYN`	0.008	*mg P/L*	Non-fixing cyanobacteria Half saturation growth for DIP
41	`KHS_O2_CYN`	0.60	*mg O2/L*	Non-fixing cyanobacteria Half saturation growth for O2
42	`FRAC_CYN_EXCR`	0.30	-	Non-fixing cyanobacteria Fraction of excretion in metabolism rate
43	`I_S_CYN`	100.00	*langleys*	Non-fixing cyanobacteria Light saturation (langleys).Not used, is calculated!
44	`DO_STR_HYPOX_CYN_D`	0.70	*mg O2/L*	Non-fixing cyanobacteria Dissolved oxygen stress in oxygen units (mortality increase below this value exponentialy
45	`THETA_HYPOX_CYN_D`	1.50	-	Non-fixing cyanobacteria Multiplier of the exponent for Dissolved oxygen stress
46	`EXPON_HYPOX_CYN_D`	1.10	*L/mg O2*	Non-fixing cyanobacteria Exponent constant for Dissolved oxygen stress
47	`CYN_N_TO_C`	0.220	*mg N/mg C*	Non-fixing cyanobacteria Nitrogen to Carbon ratio ,was 0.1
48	`CYN_P_TO_C`	0.024	*mg P/mg C*	Non-fixing cyanobacteria Phosphorus to Carbon ratio
49	`CYN_O2_TO_C`	2.66	*mg O2/mg C*	Non-fixing cyanobacteria Oxygen to Carbon ratio for respiration
50	`CYN_C_TO_CHLA`	40.00	*mg C/mg Chla*	Non-fixing cyanobacteria Carbon to Chlorophyl a ratio

Literature References:

[1] Robarts, R.D. & Zohary, T. (1987)
Temperature effects on photosynthetic capacity, respiration, and growth rates of bloom-forming cyanobacteria. New Zealand J. Marine and Freshwater Research, 21(3), 391-399.
Relevance: Optimal growth temperature for Microcystis 25-30 deg C.

[2] Reynolds, C.S. (2006)
The Ecology of Phytoplankton. Cambridge Univ. Press.
Relevance: Cyanobacteria max growth rates 1.0-3.5 /day; C:Chl-a = 30-60.

[3] Rosso, L. et al. (1993)
As above -- CTMI model for temperature response.
Relevance: T_min = 10-18 deg C, T_opt = 25-30 deg C, T_max = 35-42 deg C for cyanobacteria.

[4] Jorgensen, S.E. et al. (1991)
Handbook of Ecological Parameters and Ecotoxicology. Elsevier.
Relevance: Half-saturation constants for DIN (0.005-0.05 mg/L) and DIP (0.003-0.015 mg/L).

Nitrogen-Fixing Cyanobacteria

Parameters for heterocystous N2-fixing cyanobacteria (e.g. Aphanizomenon, Dolichospermum). Unique N-fixation parameters R_FIX and K_FIX control the transition from DIN uptake to atmospheric N2 fixation.

Value Selection Rationale:

KG_FIX_CYN_OPT_TEMP = 3.5 /day: upper range, compensating for the energetic cost of nitrogenase (Staal et al., 2003). CTMI parameters: OPT_TEMP_LR = T_min = 18, OPT_TEMP_UR = T_opt = 26, KAPPA_OVER_OPT_TEMP = T_max = 38 degC (Grimaud et al., 2017). KAPPA_UNDER_OPT_TEMP is unused. R_FIX=1.0 means non-fixing and fixing pathways balanced; K_FIX=0.008 sets switching sensitivity to low DIN (Horne & Goldman, 1994).

#	Constant Name	Value	Unit	Description
51	KG_FIX_CYN_OPT_TEMP	3.5	*1/day*	Fixing cyanobacteria Growth rate constant
52	FIX_CYN_OPT_TEMP_LR	18.0	*deg C*	Fixing Cyanobacteria CTMI T_min
53	FIX_CYN_OPT_TEMP_UR	26.0	*deg C*	Fixing Cyanobacteria CTMI T_opt
54	EFF_FIX_CYN_GROWTH	0.95	-	Fixing cyanobacteria Effective growth. (1-EG)*growth - losses for RESP and excretion
55	KAPPA_FIX_CYN_UNDER_OPT_TEMP	0.0	*deg C*	(unused)
56	KAPPA_FIX_CYN_OVER_OPT_TEMP	38.0	*deg C*	Fixing cyanobacteria CTMI T_max
57	KR_FIX_CYN_20	0.06	*1/day*	Fixing cyanobacteria RESP rate constant
58	THETA_KR_FIX_CYN	1.04	-	Fixing cyanobacteria Temperature correction for RESP rate
59	KD_FIX_CYN_20	0.10	*1/day*	Fixing cyanobacteria Mortality rate constant
60	THETA_KD_FIX_CYN	1.05	-	Fixing cyanobacteria Temperature correction for Mortality rate
61	KHS_DIN_FIX_CYN	0.01	*mg N/L*	Fixing cyanobacteria Half saturation growth for DIN
62	KHS_DIP_FIX_CYN	0.005	*mg P/L*	Fixing cyanobacteria Half saturation growth for DIP
63	KHS_O2_FIX_CYN	0.60	*mg O2/L*	Fixing cyanobacteria Half saturation growth for O2
64	FRAC_FIX_CYN_EXCR	0.30	-	Fixing cyanobacteria Fraction of excretion in metabolism rate
65	I_S_FIX_CYN	100.00	*langleys*	Fixing cyanobacteria Light saturation (langleys). Not used, is calculated!
66	DO_STR_HYPOX_FIX_CYN_D	0.70	*mg O2/L*	Fixing cyanobacteria Dissolved oxygen stress in oxygen units (mortality increase below this value exponentialy
67	THETA_HYPOX_FIX_CYN_D	1.50	-	Fixing cyanobacteria Multiplier of the exponent for Dissolved oxygen stress
68	EXPON_HYPOX_FIX_CYN_D	1.10	*L/mg O2*	Fixing cyanobacteria Exponent constant for Dissolved oxygen stress
69	FIX_CYN_N_TO_C	0.220	*mg N/mg C*	Fixing cyanobacteria Nitrogen to Carbon ratio
70	FIX_CYN_P_TO_C	0.024	*mg P/mg C*	Fixing cyanobacteria Phosphorus to Carbon ratio
71	FIX_CYN_O2_TO_C	2.66	*mg O2/mg C*	Fixing cyanobacteria Oxygen to Carbon ratio for respiration
72	FIX_CYN_C_TO_CHLA	40.00	*mg C/mg Chla*	Fixing cyanobacteria Carbon to Chlorophyl a ratio
73	R_FIX	1.0	-	Fixing cyanobacteria Ratio between non-fixing and fixing fractions growth rate
74	K_FIX	0.008	-	Fixing cyanobacteria Effectivity parameter of switching to nitrogen fixation

Literature References:

[1] Staal, M., Meysman, F.J.R. & Stal, L.J. (2003)
Temperature excludes N2-fixing heterocystous cyanobacteria in the tropical oceans. Nature, 425, 504-507.
Relevance: Temperature controls on N-fixation; optimal range 20-30 deg C.

[2] Horne, A.J. & Goldman, C.R. (1994)
Limnology, 2nd ed. McGraw-Hill.
Relevance: N-fixation rates and conditions triggering fixation when DIN is low.

[3] Paerl, H.W. & Otten, T.G. (2013)
Harmful cyanobacterial blooms: causes, consequences, and controls. Microbial Ecology, 65(4), 995-1010.
Relevance: Growth ecology and bloom dynamics of N-fixing cyanobacteria.

[4] Grimaud, G.M., Mairet, F. et al. (2017)
Modeling the temperature effect on the specific growth rate of phytoplankton: a review. Reviews in Environmental Science and Bio/Technology, 16, 625-645.
Relevance: Review of CTMI and other temperature-growth models for phytoplankton.

Other Phytoplankton (Chlorophyta, Cryptophyta, etc.)

Growth and loss parameters for a generic 'other phytoplankton' group representing green algae, cryptophytes, and chrysophytes.

Value Selection Rationale:

KG_OPA_OPT_TEMP = 2.9 /day: mid-range for green algae (Reynolds & Irish, 1997). CTMI parameters: OPT_TEMP_LR = T_min, OPT_TEMP_UR = T_opt = 20 degC, KAPPA_OVER_OPT_TEMP = T_max; KAPPA_UNDER_OPT_TEMP unused. T_opt is lower than cyanobacteria, reflecting cooler-season dominance (Sommer (Ed.), 1989). Stoichiometric ratios match Redfield. C:Chla = 30 as for diatoms.

#	Constant Name	Value	Unit	Description
75	KG_OPA_OPT_TEMP	2.9	*1/day*	OtherPhyto Growth rate constant
76	OPA_OPT_TEMP_LR	9.0	*deg C*	OtherPhyto CTMI T_min
77	OPA_OPT_TEMP_UR	20.0	*deg C*	OtherPhyto CTMI T_opt
78	EFF_OPA_GROWTH	0.95	-	OtherPhyto Effective growth. (1-EG)*growth - losses for respiration and excretion
79	KAPPA_OPA_UNDER_OPT_TEMP	0.0	*deg C*	(unused)
80	KAPPA_OPA_OVER_OPT_TEMP	33.0	*deg C*	OtherPhyto CTMI T_max
81	KR_OPA_20	0.06	*1/day*	OtherPhyto Respiration rate constant
82	THETA_KR_OPA	1.02	-	OtherPhyto Temperature correction for respiration rate
83	KD_OPA_20	0.11	*1/day*	OtherPhyto Mortality rate constant
84	THETA_KD_OPA	1.02	-	OtherPhyto Temperature correction for Mortality rate
85	KHS_DIN_OPA	0.015	*mg N/L*	OtherPhyto Half saturation growth for DIN
86	KHS_DIP_OPA	0.013	*mg P/L*	OtherPhyto Half saturation growth for DIP
87	KHS_O2_OPA	0.60	*mg O2/L*	OtherPhyto Half saturation growth for O2
88	FRAC_OPA_EXCR	0.30	-	OtherPhyto Fraction of excretion in metabolism rate
89	I_S_OPA	100.00	*langleys*	OtherPhyto Light saturation (langleys). Not used, is calculated!
90	DO_STR_HYPOX_OPA_D	0.70	*mg O2/L*	OtherPhyto Dissolved oxygen stress in oxygen units (mortality increase below this value exponentialy
91	THETA_HYPOX_OPA_D	1.20	-	OtherPhyto Multiplier of the exponent for Dissolved oxygen stress
92	EXPON_HYPOX_OPA_D	1.04	*L/mg O2*	OtherPhyto Exponent constant for Dissolved oxygen stress
93	OPA_N_TO_C	0.220	*mg N/mg C*	OtherPhyto Nitrogen to Carbon ratio
94	OPA_P_TO_C	0.024	*mg P/mg C*	OtherPhyto Phosphorus to Carbon ratio
95	OPA_O2_TO_C	2.66	*mg O2/mg C*	OtherPhyto Oxygen to Carbon ratio for respiration
96	OPA_C_TO_CHLA	30.00	*mg C/mg Chla*	OtherPhyto Carbon to Chlorophyl a ratio

Literature References:

[1] Reynolds, C.S. & Irish, A.E. (1997)
Modelling phytoplankton dynamics in lakes and reservoirs: the problem of in-situ growth rates. Hydrobiologia, 349, 5-17.
Relevance: In-situ growth rates for green algae, cryptophytes; mu_max 1.5-3.5 /day.

[2] Sommer, U. (Ed.) (1989)
Plankton Ecology: Succession in Plankton Communities. Springer.
Relevance: OtherPhyto T_opt typically 15-22 deg C (cooler than cyanobacteria). Edited volume.

Zooplankton

Growth, grazing preferences, half-saturation constants, respiration, and mortality for the bulk zooplankton compartment (mainly crustacean micro- and mesozooplankton).

Value Selection Rationale:

KG_ZOO_OPT_TEMP = 0.45 /day: within 0.2-0.8 /day (Jorgensen & Bendoricchio, 2001; Jorgensen, 1976). Grazing rate multipliers (GRAT_ZOO_*) = 1.0 for live phytoplankton, 0.5 for detritus. Preferences (PREF_ZOO_*) sum to ~1.0 with OtherPhyto highest (0.37) and cyanobacteria low (0.10), reflecting selective avoidance of cyanobacteria by Daphnia (Sommer & Sommer, 2006). Half-saturations 0.07-0.50 mg C/L (Hansen et al., 1997). FOOD_MIN_ZOO = 0.02 mg C/L prevents grazing at very low food (Jorgensen, 1976).

#	Constant Name	Value	Unit	Description
97	KG_ZOO_OPT_TEMP	0.45	*1/day*	Zooplankton Growth rate
98	ZOO_OPT_TEMP_LR	10.0	*deg C*	Zooplankton CTMI T_min
99	ZOO_OPT_TEMP_UR	25.0	*deg C*	Zooplankton CTMI T_opt
100	EFF_ZOO_GROWTH	0.80	-	Zooplankton Effective growth. (1-EG)*growth - losses for respiration and excretion
101	KAPPA_ZOO_UNDER_OPT_TEMP	0.0	*deg C*	(unused)
102	KAPPA_ZOO_OVER_OPT_TEMP	35.0	*deg C*	Zooplankton CTMI T_max
103	GRAT_ZOO_DIA	1.00	-	Zooplankton Grazing rate (growhth rate multiplier) on diatoms
104	GRAT_ZOO_CYN	1.00	-	Zooplankton Grazing rate (growhth rate multiplier) on Cyanobacteria
105	GRAT_ZOO_OPA	1.00	-	Zooplankton Grazing rate (growhth rate multiplier) on OtherPhyto
106	GRAT_ZOO_FIX_CYN	1.00	-	Zooplankton Grazing rate (growhth rate multiplier) on fixing Cyanobacteria
107	GRAT_ZOO_NOST_VEG_HET	1.00	-	Zooplankton Grazing rate (growhth rate multiplier) on Nostocles (veg + het)
108	GRAT_ZOO_DET_PART_ORG_C	0.50	-	Zooplankton Grazing rate (growhth rate multiplier) on part. ORG_C
109	PREF_ZOO_DIA	0.26	-	Zooplankton Preference for Diatoms
110	PREF_ZOO_CYN	0.10	-	Zooplankton Preference for Cyanobacteria
111	PREF_ZOO_FIX_CYN	0.07	-	Zooplankton Preference for fixing Cyanobacteria
112	PREF_ZOO_NOST_VEG_HET	0.00	-	Zooplankton Preference for nostocales (veg + het)
113	PREF_ZOO_OPA	0.37	-	Zooplankton Preference for OtherPhyto
114	PREF_ZOO_DET_PART_ORG_C	0.20	-	Zooplankton Preference for Part. ORG_C
115	KHS_DIA_C_ZOO	0.10	*mg C/L*	Zooplankton Half saturation growth for diatoms
116	KHS_CYN_C_ZOO	0.07	*mg C/L*	Zooplankton Half saturation growth for Cyanobacteria
117	KHS_FIX_CYN_C_ZOO	0.07	*mg C/L*	Zooplankton Half saturation growth for fixing Cyanobacteria
118	KHS_NOST_VEG_HET_C_ZOO	0.07	*mg C/L*	Zooplankton Half saturation growth for Nostocales (veg + het)
119	KHS_OPA_C_ZOO	0.15	*mg C/L*	Zooplankton Half saturation growth for OtherPhyto
120	KHS_DET_PART_ORG_C_ZOO	0.50	*mg C/L*	Zooplankton Half saturation growth for part. ORG_C
121	FOOD_MIN_ZOO	0.02	*mg C/L*	Zooplankton Minimum food conc. for feeding
122	KE_ZOO	0.05	-	not used Zooplankton Excretion rate as growth fraction
123	FRAC_ZOO_EX_ORG	0.30	-	not used Zooplankton Excretion rate organic fraction
124	KR_ZOO_20	0.03	*1/day*	Zooplankton Respiration rate
125	THETA_KR_ZOO	1.04	-	Zooplankton Respiration rate Temperature correction
126	KD_ZOO_20	0.15	*1/day*	Zooplankton Mortality rate
127	THETA_KD_ZOO	1.04	-	Zooplankton Mortality rate Temperature correction
128	DO_STR_HYPOX_ZOO_D	2.00	*mg O2/L*	Zooplankton Dissolved oxygen stress in oxygen units (mortality increase below this value exponentialy
129	THETA_HYPOX_ZOO_D	1.20	-	Zooplankton Multiplier of the exponent for Dissolved oxygen stress
130	EXPON_HYPOX_ZOO_D	1.06	*L/mg O2*	Zooplankton Exponent constant for Dissolved oxygen stress
131	ZOO_N_TO_C	0.220	*mg N/mg C*	Zooplankton Nitrogen to Carbon ratio

#	Constant Name	Value	Unit	Description
132	ZOO_P_TO_C	0.024	*mg P/mg C*	Zooplankton Phosphorus to Carbon ratio
133	ZOO_O2_TO_C	2.66	*mg O2/mg C*	Zooplankton Oxygen to Carbon ratio for respiration

Literature References:

[1] Jorgensen, S.E. & Bendoricchio, G. (2001)
Fundamentals of Ecological Modelling, 3rd ed. Elsevier.
Relevance: Zooplankton growth rates 0.2-0.8 /day; model uses 0.45 /day.

[2] Hansen, P.J., Bjornsen, P.K. & Hansen, B.W. (1997)
Zooplankton grazing and growth: scaling within the 2-2000 um body size range. Limnology and Oceanography, 42, 687-704.
Relevance: Grazing rates scaling with body size; half-saturation for food 0.02-0.5 mg C/L.

[3] Jorgensen, S.E. (1976)
A eutrophication model for a lake. Ecological Modelling, 2(2), 147-165.
Relevance: Threshold food concentration for zooplankton grazing (FOOD_MIN ~ 0.02 mg C/L). Zooplankton N:C ~ 0.17-0.25, P:C ~ 0.015-0.03.

[4] Sommer, U. & Sommer, F. (2006)
Cladocerans versus copepods: the cause of contrasting top-down controls on freshwater and marine phytoplankton. Oecologia, 147, 183-194.
Relevance: Selective feeding preferences in freshwater zooplankton communities.

Particulate Organic Matter (POM) Dissolution

First-order dissolution rates for particulate organic carbon (POC), nitrogen (PON), phosphorus (POP), and biogenic silica (BSi). Phytoplankton-enhancement factors simulate bacterial colonisation.

Value Selection Rationale:

KDISS POC = 10.0 /day: high value accounts for rapid labile fraction; PON = 0.25 /day and POP = 3.48 /day within Wetzel (2001) ranges (PON 0.05-0.5, POP 0.1-5.0 /day). BSi dissolution = 0.001 /day is low, consistent with Kamatani (1982). Theta = 1.06 for temperature correction (Bowie et al., 1985). Phytoplankton factors near zero disable enzymatic enhancement for PON/POP to avoid double counting.

#	Constant Name	Value	Unit	Description
134	KDISS_DET_PART_ORG_C_20	10.0	*1/day*	POC Dissolution rate not dependent on phytoplankton
135	THETA_KDISS_DET_PART_ORG_C	1.06	-	POC Carbon Dissolution rate Temperature correction
136	FAC_PHYT_DET_PART_ORG_C	2.	*L/(mg C)*	!0.001 POC Carbon Phytoplankton linear factor for dissolution rate
137	KDISS_DET_PART_ORG_N_20	0.25	*1/day*	PON Dissolution rate not dependent on phytoplankton
138	THETA_KDISS_DET_PART_ORG_N	1.06	-	PON Dissolution rate Temperature correction
139	KHS_DISS_N	0.025	*mg N/L*	PON dissolution reverse half saturation for DIN
140	FAC_PHYT_DET_PART_ORG_N	0.	*L/(mg C)*	!0.007 PON Phytoplankton linear factor for dissolution rate
141	KDISS_DET_PART_ORG_P_20	3.48	*1/day*	POP Phosphorus Dissolution rate not dependent on phytoplankton
142	THETA_KDISS_DET_PART_ORG_P	1.06	-	POP Dissolution rate Temperature correction
143	KHS_DISS_P	0.03	*mg P/L*	POP Dissolution reverse half saturation for DIP
144	FAC_PHYT_DET_PART_ORG_P	0.	*L/(mg C)*	!0.005 POP Phytoplankton linear factor for dissolution rate
145	KDISS_PART_Si_20	0.001	*1/day*	Particulate Silica Dissolution rate
146	THETA_KDISS_PART_Si	1.04	-	Particulate Silica Dissolution rate Temperature correction

Literature References:

[1] Wetzel, R.G. (2001)
Limnology: Lake and River Ecosystems, 3rd ed. Academic Press.
Relevance: POM dissolution rates: POC 0.01-0.1 /day; PON 0.05-0.5 /day; POP 0.1-1.0 /day.

[2] Bowie, G.L. et al. (1985)
As above -- EPA compilation.
Relevance: Temperature correction theta 1.04-1.08 for organic matter dissolution.

[3] Kamatani, A. (1982)
Dissolution rates of silica from diatoms decomposing at various temperatures. Marine Biology, 68, 91-96.
Relevance: Biogenic silica dissolution rate 0.001-0.01 /day, increasing with temperature.

Dissolved Organic Matter (DOM) Mineralisation

Phytoplankton-dependent mineralisation rate factors and reverse half-saturation concentrations for DOC, DON, and DOP.

Value Selection Rationale:

FAC_PHYT_AMIN_DOC = 0.0045: low value reflects slow co-metabolic mineralisation. KHS_AMIN_N = 100 mg/L effectively disables the reverse half-saturation for DON (labile pool always mineralized). DOP factor = 0.90 reflects rapid enzymatic DOP hydrolysis by alkaline phosphatase (Chapra, 1997).

#	Constant Name	Value	Unit	Description
147	FAC_PHYT_AMIN_DOC	0.0045	*L/(mg C)*	0.045 DOC Phytoplankton linear factor for mineralisation rate
148	KHS_AMIN_N	100.	*mg/L*	DON mineralisation reverse half saturation for DIN
149	FAC_PHYT_AMIN_DON	0.008	*L/(mg C)*	DON Phytoplankton linear factor for mineralisation rate
150	KHS_AMIN_P	0.025	*mg/L*	DOP reverse half saturation for DIP
151	FAC_PHYT_AMIN_DOP	0.90	*L/(mg C)*	DOP Phytoplankton linear factor for mineralisation rate

Literature References:

[1] Chapra, S.C. (1997)
Surface Water-Quality Modeling. McGraw-Hill.
Relevance: DOM mineralisation rates and phytoplankton enhancement factors.

[2] Bowie, G.L. et al. (1985)
As above.
Relevance: DOC mineralisation rates 0.01-0.2 /day; DON 0.005-0.1 /day.

Nitrification

Ammonia oxidation (nitrification) rate, temperature correction, half-saturation for O2 and NH4-N, and optimum pH range.

Value Selection Rationale:

K_NITR_20 = 0.6 /day: within 0.1-1.0 /day (Chapra, 1997; Cerco, 2000). THETA = 1.045 standard (Bowie et al., 1985). KHS_NITR_OXY = 2.0 mg/L: nitrification inhibited below ~2 mg O2/L (Painter, 1970). pH range 6.9-8.2 is the active window (Painter, 1970).

#	Constant Name	Value	Unit	Description
152	`K_NITR_20`	0.600	*1/day*	Amonia nitrification rate
153	`THETA_K_NITR`	1.045	-	Amonia nitrification rate Temperature constant
154	`KHS_NITR_OXY`	2.000	*mg O2/L*	Amonia nitrification half saturation for Oxygen
155	`KHS_NITR_NH4_N`	0.030	*mg N/L*	Amonia nitrification half saturation for Amonia
156	`PH_NITR_NH4_MIN`	6.90	*pH*	optimum lower range for pH correction factor for nitrification
157	`PH_NITR_NH4_MAX`	8.20	*pH*	optimum upper range for pH correction factor for nitrification

Literature References:

[1] Chapra, S.C. (1997)
Surface Water-Quality Modeling. McGraw-Hill.
Relevance: Nitrification rate 0.1-1.0 /day at 20 deg C; theta = 1.04-1.08.

[2] Bowie, G.L. et al. (1985)
As above.
Relevance: KHS_O2 for nitrification 0.5-4.0 mg/L; KHS_NH4 0.01-0.10 mg N/L.

[3] Painter, H.A. (1970)
A review of literature on inorganic nitrogen metabolism in microorganisms. Water Research, 4(6), 393-450.
Relevance: pH optimum range for nitrification 6.5-8.5; strong inhibition below pH 6.0.

[4] Cerco, C.F. (2000)
Phytoplankton kinetics in the Chesapeake Bay eutrophication model. Water Quality and Ecosystems Modeling, 1, 5-49.
Relevance: Nitrification parameters used in major estuary models.

Redox Metal Chemistry (Fe, Mn)

Oxidation/reduction rate constants for Fe(II)/Fe(III) and Mn(II)/Mn(IV), plus reversed Monod half-saturation for dissolved oxygen inhibition.

Value Selection Rationale:

k_OX_FE_II = 0.00125 /day: slow consistent with Stumm & Morgan (1996) range 10^-3 to 10^-1 /day. k_RED rates = 2.0 /day: within 0.5-5.0 (Davison, 1993). KHS_DOXY thresholds = 0.20 mg/L: reduction proceeds only under near-anoxic conditions.

#	Constant Name	Value	Unit	Description
158	`k_OX_FE_II`	0.00125	*1/day*	Oxidation rate for iron 2+
159	`k_RED_FE_III`	2.00	*1/day*	reduction rate for iron 3+
160	`k_OX_MN_II`	0.01	*1/day*	oxidation rate for manganese 2+
161	`k_RED_MN_IV`	2.00	*1/day*	reduction rate for manganese 4+
162	`KHS_DOXY_FE_III_RED`	0.20	*mg O2/L*	reversed Monod half saturation of DOXY for iron 3+ reduction
163	`KHS_DOXY_MN_IV_RED`	0.20	*mg O2/L*	reversed Monod half saturation of DOXY for manganese 4+ reduction

Literature References:

[1] Stumm, W. & Morgan, J.J. (1996)
Aquatic Chemistry: Chemical Equilibria and Rates in Natural Waters, 3rd ed. Wiley.
Relevance: Fe(II) oxidation rate 10^-3 to 10^-1 /day; Mn(II) oxidation 10^-2 to 10^-1 /day.

[2] Davison, W. (1993)
Iron and manganese in lakes. Earth-Science Reviews, 34(2), 119-163.
Relevance: Reduction rates of Fe(III) and Mn(IV) under anoxic conditions 0.5-5.0 /day.

DOC Mineralisation -- Multi-Electron-Acceptor

Mineralisation rate constants of DOC using different terminal electron acceptors (O2, NO3, Mn(IV), Fe(III), SO4, methanogenesis). Includes temperature corrections, half-saturations, inhibition following the thermodynamic sequence, and pH optima.

Value Selection Rationale:

K_MIN_DOC_DOXY_20 = 0.010 /day: aerobic rate intentionally low to represent semi-refractory DOC pool; anaerobic rates = 0.025 /day for all alternative acceptors, following Van Cappellen & Wang (1996). Theta = 1.04 throughout (Soetaert et al., 1996). Inhibition KHS values = 0.10 mg/L enforce the thermodynamic sequence O2 > NO3 > Mn(IV) > Fe(III) > SO4 > methanogenesis (Berner, 1980). pH optima 6.0-9.0 (Boudreau, 1997).

#	Constant Name	Value	Unit	Description
164	`K_MIN_DOC_DOXY_20`	0.010	*1/day*	Mineralization rate constant of DOC at 20 C for dissolved oxygen as final electron acceptor
165	`K_MIN_DOC_NO3N_20`	0.025	*1/day*	Mineralization rate constant of DOC at 20 C for nitrate as final electron acceptor
166	`K_MIN_DOC_MN_IV_20`	0.025	*1/day*	Mineralization rate constant of DOC at 20 C for Mn IV as final electron acceptor
167	`K_MIN_DOC_FE_III_20`	0.025	*1/day*	Mineralization rate constant of DOC at 20 C for Fe III as final electron acceptor
168	`K_MIN_DOC_S_PLUS_6_20`	0.025	*1/day*	Mineralization rate constant of DOC at 20 C for sulphate as final electron acceptor
169	`K_MIN_DOC_DOC_20`	0.025	*1/day*	Mineralization rate constant of DOC at 20 C for DOC itself as final electron acceptor
170	`THETA_K_MIN_DOC_DOXY`	1.04	-	Temperature correction factor for mineralization process rate constant of DOC for dissolved oxygen as final electron acceptor
171	`THETA_K_MIN_DOC_NO3N`	1.04	-	Temperature correction factor for mineralization process rate constant of DOC for nitrate as final electron acceptor
172	`THETA_K_MIN_DOC_MN_IV`	1.04	-	Temperature correction factor for mineralization process rate constant of DOC for MN IV as final electron acceptor
173	`THETA_K_MIN_DOC_FE_III`	1.04	-	Temperature correction factor for mineralization process rate constant of DOC for FE III as final electron acceptor
174	`THETA_K_MIN_DOC_S_PLUS_6`	1.04	-	Temperature correction factor for mineralization process rate constant of DOC for Suphate as final electron acceptor
175	`THETA_K_MIN_DOC_DOC`	1.04	-	Temperature correction factor for mineralization process rate constant of DOC for DOC
176	`K_HS_DOC_MIN_DOXY`	0.00	*mg O2/L*	Monod type half-saturation concentration of DOC for DOC mineralization for dissolved oxygen as the final electron acceptor
177	`K_HS_DOC_MIN_NO3N`	1.00	*mg N/L*	Monod type half-saturation concentration of DOC for DOC mineralization for nitrate as the final electron acceptor
178	`K_HS_DOC_MIN_MN_IV`	1.00	*mg C/L*	Monod type half-saturation concentration of DOC for DOC mineralization for Mn IV as the final electron acceptor
179	`K_HS_DOC_MIN_FE_III`	1.00	*mg C/L*	Monod type half-saturation concentration of DOC for DOC mineralization for Fe III as the final electron acceptor
180	`K_HS_DOC_MIN_S_PLUS_6`	1.00	*mg C/L*	Monod type half-saturation concentration of DOC for DOC mineralization for sulphate as the final electron acceptor
181	`K_HS_DOC_MIN_DOC`	1.00	*mg C/L*	Monod type half-saturation concentration of DOC for DOC mineralization for DOC itself as the final electron acceptor
182	`K_HS_DOXY_RED_LIM`	1.00	*mg O2/L*	Monod type half-saturation concentration of dissolved oxygen to limit the consumption of dissolved oxygen as the final electron acce
183	`K_HS_NO3N_RED_LIM`	1.00	*mg N/L*	Monod type half-saturation concentration of nitrate nitrogen to limit the consumption of nitrate as the final electron acceptor for
184	`K_HS_MN_IV_RED_LIM`	1.00	*mg Mn/L*	Monod type half-saturation concentration of MN_IV to limit the consumption of Mn IV as the final electron acceptor for DOC mineraliz
185	`K_HS_FE_III_RED_LIM`	1.00	*mg Fe/L*	Monod type half-saturation concentration of FE_III to limit the consumption of Fe III as the final electron acceptor for DOC mineral
186	`K_HS_S_PLUS_6_RED_LIM`	1.00	*mg S/L*	Monod type half-saturation concentration of sulphate to limit the consumption of sulphate as the final electron acceptor for DOC min
187	`K_HS_DOXY_RED_INHB`	0.10	*mg O2/L*	Reversed Monod type half saturation concentration that simulate the inhibition effect of dissolved oxygen on consumption of NO3, Mn
188	`K_HS_NO3N_RED_INHB`	0.10	*mg N/L*	Reversed Monod type half saturation concentration that simulate the inhibition effect of nitrate on consumption of Mn IV, Fe III, S

#	Constant Name	Value	Unit	Description
189	`K_HS_MN_IV_RED_INHB`	0.10	*mg Mn/L*	Reversed Monod type half saturation concentration that simulate the inhibition effect of Mn IV on consumption of Fe III, S VI and DO
190	`K_HS_FE_III_RED_INHB`	0.10	*mg Fe/L*	Reversed Monod type half saturation concentration that simulate the inhibition effect of Fe III on consumption of S VI and DOC itsel
191	`K_HS_S_PLUS_6_RED_INHB`	0.10	*mg S/L*	Reversed Monod type half saturation concentration that simulate the inhibition effect of S VI on consumption of DOC itself for DOC m
192	`PH_MIN_DOC_MIN_DOXY`	6.00	*pH*	Min. pH for the optimum pH range for DOC mineralization with DOXY as final electron acceptor
193	`PH_MIN_DOC_MIN_NO3N`	6.00	*pH*	Min. pH for the optimum pH range for DOC mineralization with NO3N as final electron acceptor
194	`PH_MIN_DOC_MIN_MN_IV`	6.00	*pH*	Min. pH for the optimum pH range for DOC mineralization with MN_IV as final electron acceptor
195	`PH_MIN_DOC_MIN_FE_III`	6.00	*pH*	Min. pH for the optimum pH range for DOC mineralization with FE_III as final electron acceptor
196	`PH_MIN_DOC_MIN_S_PLUS_6`	6.00	*pH*	Min. pH for the optimum pH range for DOC mineralization with S_PLUS_6 as final electron acceptor
197	`PH_MIN_DOC_MIN_DOC`	6.00	*pH*	Min. pH for the optimum pH range for DOC mineralization with DOC as final electron acceptor
198	`PH_MAX_DOC_MIN_DOXY`	9.00	*pH*	Max. pH for the optimum pH range for DOC mineralization with dissolved oxygen as final electron acceptor
199	`PH_MAX_DOC_MIN_NO3N`	9.00	*pH*	Max. pH for the optimum pH range for DOC mineralization with NO3N as final electron acceptor
200	`PH_MAX_DOC_MIN_MN_IV`	9.00	*pH*	Max. pH for the optimum pH range for DOC mineralization with MN_IV as final electron acceptor
201	`PH_MAX_DOC_MIN_FE_III`	9.00	*pH*	Max. pH for the optimum pH range for DOC mineralization with FE_III as final electron acceptor
202	`PH_MAX_DOC_MIN_S_PLUS_6`	9.00	*pH*	Max. pH for the optimum pH range for DOC mineralization with S_PLUS_6 as final electron acceptor
203	`PH_MAX_DOC_MIN_DOC`	9.00	*pH*	Max. pH for the optimum pH range for DOC mineralization with DOC as final electron acceptor

Literature References:

[1] Van Cappellen, P. & Wang, Y. (1996)
Cycling of iron and manganese in surface sediments: a general theory for the coupled transport and reaction of carbon, oxygen, nitrogen, sulfur, iron, and manganese. American Journal of Science, 296, 197-243.
Relevance: Thermodynamic sequence of electron acceptors: O2 > NO3 > Mn(IV) > Fe(III) > SO4 > CH4. Inhibition half-saturation constants 0.01-1.0 mg/L.

[2] Berner, R.A. (1980)
Early Diagenesis: A Theoretical Approach. Princeton Univ. Press.
Relevance: Foundation for multi-electron-acceptor organic matter degradation kinetics.

[3] Soetaert, K., Herman, P.M.J. & Middelburg, J.J. (1996)
A model of early diagenetic processes from the shelf to abyssal depths. Geochimica et Cosmochimica Acta, 60(6), 1019-1040.
Relevance: DOC mineralisation rates 0.01-0.10 /day; theta 1.02-1.08.

[4] Boudreau, B.P. (1997)
Diagenetic Models and Their Implementation. Springer.
Relevance: pH correction factors for organic matter mineralisation (pH 6-9 optimum).

DON Mineralisation -- Multi-Electron-Acceptor

Analogous to DOC mineralisation but for dissolved organic nitrogen. Same multi-electron-acceptor framework.

Value Selection Rationale:

K_MIN_DON_DOXY_20 = 0.10 /day: faster than DOC aerobic rate because N-containing substrates are more labile (Soetaert et al., 1996). Anaerobic rates = 0.0012 /day. Theta = 1.08 (higher temperature sensitivity). pH optima 6.0-9.0.

#	Constant Name	Value	Unit	Description
204	K_MIN_DON_DOXY_20	0.100	*1/day*	Mineralization rate constant of DON at 20 C for dissolved oxygen as final electron acceptor
205	K_MIN_DON_NO3N_20	0.0012	*1/day*	Mineralization rate constant of DON at 20 C for NO3_N as final electron acceptor
206	K_MIN_DON_MN_IV_20	0.0012	*1/day*	Mineralization rate constant of DON at 20 C for Mn IV as final electron acceptor
207	K_MIN_DON_FE_III_20	0.0012	*1/day*	Mineralization rate constant of DON at 20 C for FE III as final electron acceptor
208	K_MIN_DON_S_PLUS_6_20	0.0012	*1/day*	Mineralization rate constant of DON at 20 C for S_PLUS_6 as final electron acceptor
209	K_MIN_DON_DOC_20	0.0012	*1/day*	Mineralization rate constant of DON at 20 C for DOC as final electron acceptor
210	THETA_K_MIN_DON_DOXY	1.08	-	Temperature correction factor for mineralization process rate constant of DON for dissolved oxygen as final electron acceptor
211	THETA_K_MIN_DON_NO3N	1.08	-	Temperature correction factor for mineralization process rate constant of DON for nitrate as final electron acceptor
212	THETA_K_MIN_DON_MN_IV	1.08	-	Temperature correction factor for mineralization process rate constant of DON for MN_IV as final electron acceptor
213	THETA_K_MIN_DON_FE_III	1.08	-	Temperature correction factor for mineralization process rate constant of DON for FE_III as final electron acceptor
214	THETA_K_MIN_DON_S_PLUS_6	1.08	-	Temperature correction factor for mineralization process rate constant of DON for sulphate as final electron acceptor
215	THETA_K_MIN_DON_DOC	1.08	-	Temperature correction factor for mineralization process rate constant of DON for DOC
216	K_HS_DON_MIN_DOXY	0.00	*mg O2/L*	Monod type half-saturation concentration of DON for DON mineralization for dissolved oxygen as the final electron acceptor
217	K_HS_DON_MIN_NO3N	0.05	*mg N/L*	Monod type half-saturation concentration of DON for DON mineralization for nitrate nitrogen as the final electron acceptor
218	K_HS_DON_MIN_MN_IV	0.05	*mg N/L*	Monod type half-saturation concentration of DON for DON mineralization for MN_IV as the final electron acceptor
219	K_HS_DON_MIN_FE_III	0.05	*mg N/L*	Monod type half-saturation concentration of DON for DON mineralization for FE_III as the final electron acceptor
220	K_HS_DON_MIN_S_PLUS_6	0.05	*mg N/L*	Monod type half-saturation concentration of DON for DON mineralization for sulphate sulphur as the final electron acceptor
221	K_HS_DON_MIN_DOC	0.05	*mg N/L*	Monod type half-saturation concentration of DON for DON mineralization for DOC as the final electron acceptor
222	PH_MIN_DON_MIN_DOXY	6.00	*pH*	Min. pH for the optimum pH range for DON mineralization with DOXY as final electron acceptor
223	PH_MIN_DON_MIN_NO3N	6.00	*pH*	Min. pH for the optimum pH range for DON mineralization with NO3_N as final electron acceptor
224	PH_MIN_DON_MIN_MN_IV	6.00	*pH*	Min. pH for the optimum pH range for DON mineralization with MN IV as final electron acceptor
225	PH_MIN_DON_MIN_FE_III	6.00	*pH*	Min. pH for the optimum pH range for DON mineralization with FE III as final electron acceptor
226	PH_MIN_DON_MIN_S_PLUS_6	6.00	*pH*	Min. pH for the optimum pH range for DON mineralization with S_PLUS_6 as final electron acceptor
227	PH_MIN_DON_MIN_DOC	6.00	*pH*	Min. pH for the optimum pH range for DON mineralization with DOC as final electron acceptor
228	PH_MAX_DON_MIN_DOXY	9.00	*pH*	Max. pH for the optimum pH range for DON mineralization with dissolved oxygen as final electron acceptor
229	PH_MAX_DON_MIN_NO3N	9.00	*pH*	Max. pH for the optimum pH range for DON mineralization with NO3_N as final electron acceptor

#	Constant Name	Value	Unit	Description
230	PH_MAX_DON_MIN_MN_IV	9.00	*pH*	Max. pH for the optimum pH range for DON mineralization with Mn IV as final electron acceptor
231	PH_MAX_DON_MIN_FE_III	9.00	*pH*	Max. pH for the optimum pH range for DON mineralization with FE_III as final electron acceptor
232	PH_MAX_DON_MIN_S_PLUS_6	9.00	*pH*	Max. pH for the optimum pH range for DON mineralization with S_PLUS_6 as final electron acceptor
233	PH_MAX_DON_MIN_DOC	9.00	*pH*	Max. pH for the optimum pH range for DON mineralization with DOC as final electron acceptor

Literature References:

[1] Soetaert, K. et al. (1996)

As above.

Relevance: DON mineralisation rates generally lower than DOC; 0.001-0.1 /day.

[2] Berner, R.A. (1980)

As above.

Relevance: Multi-electron-acceptor framework applies equally to DON.

DOP Mineralisation -- Multi-Electron-Acceptor

Analogous to DOC mineralisation but for dissolved organic phosphorus.

Value Selection Rationale:

K_MIN_DOP_DOXY_20 = 0.70 /day: DOP mineralisation is the fastest of the three DOM pools due to efficient enzymatic hydrolysis by alkaline phosphatase (Soetaert et al., 1996). Anaerobic rates = 0.03 /day.

#	Constant Name	Value	Unit	Description
234	K_MIN_DOP_DOXY_20	0.70	*1/day*	Mineralization rate constant of DOP at 20 C for dissolved oxygen as final electron acceptor
235	K_MIN_DOP_NO3N_20	0.03	*1/day*	Mineralization rate constant of DOP at 20 C for NO3_N as final electron acceptor
236	K_MIN_DOP_MN_IV_20	0.03	*1/day*	Mineralization rate constant of DOP at 20 C for Mn IV as final electron acceptor
237	K_MIN_DOP_FE_III_20	0.03	*1/day*	Mineralization rate constant of DOP at 20 C for FE_III as final electron acceptor
238	K_MIN_DOP_S_PLUS_6_20	0.03	*1/day*	Mineralization rate constant of DOP at 20 C for S_PLUS_6 as final electron acceptor
239	K_MIN_DOP_DOC_20	0.03	*1/day*	Mineralization rate constant of DOP at 20 C for DOC as final electron acceptor
240	THETA_K_MIN_DOP_DOXY	1.06	-	Temperature correction factor for mineralization process rate constant of DOP for dissolved oxygen as final electron acceptor
241	THETA_K_MIN_DOP_NO3N	1.04	-	Temperature correction factor for mineralization process rate constant of DOP for nitrate as final electron acceptor
242	THETA_K_MIN_DOP_MN_IV	1.04	-	Temperature correction factor for mineralization process rate constant of DOP for Mn IV as final electron acceptor
243	THETA_K_MIN_DOP_FE_III	1.04	-	Temperature correction factor for mineralization process rate constant of DOP for FE_III as final electron acceptor
244	THETA_K_MIN_DOP_S_PLUS_6	1.04	-	Temperature correction factor for mineralization process rate constant of DOP for sulphate as final electron acceptor
245	THETA_K_MIN_DOP_DOC	1.04	-	Temperature correction factor for mineralization process rate constant of DOP for DOC
246	K_HS_DOP_MIN_DOXY	0.00	*mg O2/L*	Monod type half-saturation concentration of DOP for DOP mineralization for dissolved oxygen as the final electron acceptor
247	K_HS_DOP_MIN_NO3N	0.052	*mg N/L*	Monod type half-saturation concentration of DOP for DOP mineralization for nitrate nitrogen as the final electron acceptor
248	K_HS_DOP_MIN_MN_IV	0.052	*mg P/L*	Monod type half-saturation concentration of DOP for DOP mineralization for MN_IV as the final electron acceptor
249	K_HS_DOP_MIN_FE_III	0.052	*mg P/L*	Monod type half-saturation concentration of DOP for DOP mineralization for FE_III as the final electron acceptor
250	K_HS_DOP_MIN_S_PLUS_6	0.052	*mg P/L*	Monod type half-saturation concentration of DOP for DOP mineralization for sulphate sulphur as the final electron acceptor
251	K_HS_DOP_MIN_DOC	0.052	*mg P/L*	Monod type half-saturation concentration of DOP for DOP mineralization for DOC as the final electron acceptor
252	PH_MIN_DOP_MIN_DOXY	6.00	*pH*	Min. pH for the optimum pH range for DOP mineralization with dissolved oxygen as final electron acceptor
253	PH_MIN_DOP_MIN_NO3N	6.00	*pH*	Min. pH for the optimum pH range for DOP mineralization with NO3_N as final electron acceptor
254	PH_MIN_DOP_MIN_MN_IV	6.00	*pH*	Min. pH for the optimum pH range for DOP mineralization with Mn IV as final electron acceptor
255	PH_MIN_DOP_MIN_FE_III	6.00	*pH*	Min. pH for the optimum pH range for DOP mineralization with FE III as final electron acceptor
256	PH_MIN_DOP_MIN_S_PLUS_6	6.00	*pH*	Min. pH for the optimum pH range for DOP mineralization with S_PLUS_6 as final electron acceptor
257	PH_MIN_DOP_MIN_DOC	6.00	*pH*	Min. pH for the optimum pH range for DOP mineralization with DOC as final electron acceptor
258	PH_MAX_DOP_MIN_DOXY	9.00	*pH*	Max. pH for the optimum pH range for DOP mineralization with dissolved oxygen as final electron acceptor
259	PH_MAX_DOP_MIN_NO3N	9.00	*pH*	Max. pH for the optimum pH range for DOP mineralization with NO3_N as final electron acceptor

#	Constant Name	Value	Unit	Description
260	PH_MAX_DOP_MIN_MN_IV	9.00	*pH*	Max. pH for the optimum pH range for DOP mineralization with Mn IV as final electron acceptor
261	PH_MAX_DOP_MIN_FE_III	9.00	*pH*	Max. pH for the optimum pH range for DOP mineralization with FE III as final electron acceptor
262	PH_MAX_DOP_MIN_S_PLUS_6	9.00	*pH*	Max. pH for the optimum pH range for DOP mineralization with S_PLUS_6 as final electron acceptor
263	PH_MAX_DOP_MIN_DOC	9.00	*pH*	Max. pH for the optimum pH range for DOP mineralization with DOC as final electron acceptor

Literature References:

[1] Soetaert, K. et al. (1996)
As above.
Relevance: DOP mineralisation typically faster than DON due to enzymatic hydrolysis.

[2] Berner, R.A. (1980)
As above.
Relevance: Phosphorus regeneration coupled to redox chemistry.

Methane & Hydrogen Sulphide

Oxidation rates for CH4 and H2S with Arrhenius temperature corrections and half-saturation for dissolved oxygen.

Value Selection Rationale:

k_OX_CH4 = 1.0 /day and k_OX_H2S = 1.0 /day: mid-range values from Bastviken et al. (2004) and Jorgensen (1982). Theta = 1.04 standard. Half-saturation for O2 = 1.5 mg/L indicates oxidation requires moderate oxygen availability.

#	Constant Name	Value	Unit	Description
264	`k_OX_CH4`	1.0	*1/day*	Methane oxidation rate constant at 20 C
265	`THETA_k_OX_CH4`	1.04	-	Arhenius type temperature correction for methane oxidation rate constant
266	`k_HS_OX_CH4_DOXY`	1.5	*mg O2/L*	Monod type Half saturation of methane oxidation for dissolved oxygen
267	`k_OX_H2S`	1.0	*1/day*	Hydrogen sulphide oxidation rate constant at 20 C
268	`THETA_k_OX_H2S`	1.04	-	Arhenius type temperature correction for sulphide oxidation rate constant
269	`k_HS_OX_H2S_DOXY`	1.5	*mg O2/L*	Monod type Half saturation of hydrogen sulphide oxidation for dissolved oxygen

Literature References:

[1] Bastviken, D. et al. (2004)
Methane emissions from lakes: dependence of lake characteristics, two regional assessments, and a global estimate. Global Biogeochemical Cycles, 18(4), GB4009.
Relevance: Methane oxidation rates 0.1-10 /day; dependent on O2 availability.

[2] Jorgensen, B.B. (1982)
Mineralization of organic matter in the sea bed -- the role of sulphate reduction. Nature, 296, 643-645.
Relevance: H2S oxidation rates in aquatic systems; 0.1-5.0 /day.

Fe Dissolution & Fractionation

Dissolution rate constants for particulate Fe(II) and Fe(III) phases and initial dissolved fractions.

Value Selection Rationale:

k_DISS_FE_II_20 = 0.1 /day and k_DISS_FE_III_20 = 0.10 /day: within 0.01-1.0 /day range (Stumm & Morgan, 1996). Initial dissolved Fe(II) fraction = 0.01 (mostly particulate); Fe(III) fraction = 0.50 (equilibrium partitioning).

#	Constant Name	Value	Unit	Description
270	`k_DISS_FE_II_20`	0.1	-	Dissolution rate constant for Fe II at 20 C
271	`THETA_k_DISS_FE_II`	1.04	-	Dissolution rate constant for Fe II
272	`INIT_MULT_FE_II_DISS`	0.01	-	Initial fraction of dissolved Fe II
273	`k_DISS_FE_III_20`	0.10	-	Dissolution rate constant for Fe III at 20 C
274	`THETA_k_DISS_FE_III`	1.04	-	Dissolution rate constant for Fe III
275	`INIT_MULT_FE_III_DISS`	0.50	-	Initial fraction of dissolved Fe III

Literature References:

[1] Stumm, W. & Morgan, J.J. (1996)

As above.

Relevance: Dissolved vs. particulate Fe fractionation; dissolution rates 0.01-1.0 /day.

Nostocales (Heterocystous Cyanobacteria)

Growth, loss, and akinete (resting stage) formation/germination parameters for Nostocales (Dolichospermum, Aphanizomenon flos-aquae). Includes density-dependent mortality and seasonal triggers.

Value Selection Rationale:

KG_NOST_VEG_HET = 1.29 /day within 0.5-2.0 /day (Paerl & Otten, 2013). T_opt=26, T_max=38 degC (Sukenik et al., 2012; Kovacs et al., 2016). Akinete germination rate P_GERM_AKI = 0.3 /day, triggered when T > 21 degC AND DIN < 0.1 mg/L (Hense & Beckmann, 2006). Formation starts after day 200 when T < 16 degC.

#	Constant Name	Value	Unit	Description
276	KG_NOST_VEG_HET_OPT_TEMP	1.29	*1/day*	Nostocales (veg + het) Growth rate constant
277	NOST_VEG_HET_OPT_TEMP_LR	16.0	*deg C*	Nostocales (veg + het) CTMI T_min
278	NOST_VEG_HET_OPT_TEMP_UR	26.0	*deg C*	Nostocales (veg + het) CTMI T_opt
279	EFF_NOST_VEG_HET_GROWTH	0.95	-	Nostocales (veg + het) Effective growth. (1-EG)*growth - losses for RESP and excretion
280	KAPPA_NOST_VEG_HET_UNDER_OPT_TEMP	0.0	*deg C*	(unused)
281	KAPPA_NOST_VEG_HET_OVER_OPT_TEMP	38.0	*deg C*	Nostocales (veg + het) CTMI T_max
282	KR_NOST_VEG_HET_20	0.06	*1/day*	Nostocales (veg + het) RESP rate constant
283	THETA_KR_NOST_VEG_HET	1.04	-	Nostocales (veg + het) Temperature correction for RESP rate
284	KD_NOST_VEG_HET_20	0.040	*1/day*	Nostocales (veg + het) Mortality rate constant
285	THETA_KD_NOST_VEG_HET	1.05	-	Nostocales (veg + het) Temperature correction for Mortality rate
286	M_DENS_VEG_HET	0.001	*L/(mg C day)*	Nostocales (veg + het) mortality rate constant because of too dense population
287	KHS_DP_NOST_VEG_HET	0.005	*mg P/L*	Nostocales (veg + het) Half saturation growth for DP
288	KHS_O2_NOST_VEG_HET	0.60	*mg O2/L*	Nostocales (veg + het) Half saturation growth for O2
289	FRAC_NOST_VEG_HET_EXCR	0.30	-	Nostocales (veg + het) Fraction of excretion in metabolism rate
290	I_S_NOST_VEG_HET	100.	*langleys*	Nostocales (veg + het) Light saturation (langleys). Not used, is calculated!
291	DO_STR_HYPOX_NOST_VEG_HET_D	0.70	*mg O2/L*	Nostocales (veg + het) Dissolved oxygen stress in oxygen units (mortality increase below this value exponentialy
292	THETA_HYPOX_NOST_VEG_HET_D	1.50	-	Nostocales (veg + het) Multiplier of the exponent for Dissolved oxygen stress
293	EXPON_HYPOX_NOST_VEG_HET_D	1.10	*L/mg O2*	Nostocales (veg + het) Exponent constant for Dissolved oxygen stress
294	NOST_N_TO_C	0.220	*mg N/mg C*	Nostocales (veg + het) Nitrogen to Carbon ratio
295	NOST_P_TO_C	0.024	*mg P/mg C*	Nostocales (veg + het) Phosphorus to Carbon ratio
296	NOST_O2_TO_C	2.66	*mg O2/mg C*	Nostocales (veg + het) Oxygen to Carbon ratio for respiration
297	NOST_C_TO_CHLA	40.00	*mg C/mg Chla*	Nostocales (veg + het) Carbon to Chlorophyl a ratio
298	P_GERM_AKI	0.3	*1/day*	Germination rate constant for akinets, 1/day
299	N_GERM_AKI	0.1	*mg N/L*	DIN concentration to trigger germination (below that germination starts)
300	P_FORM_AKI	0.1	*1/day*	Formation rate constant for akinets
301	DAY_FORM_AKI	200	*day of year*	Day of the year from which formation of akinetes is allowed
302	T_FORM_AKI	16.0	*deg C*	Temperature to trigger formation (below that formation starts)
303	K_LOSS_AKI	0.000	*1/day*	Loss rate constant for akinetes
304	K_MORT_AKI_20	0.000	*1/day*	Mortality rate constant of akinetes at 20 degrees celsius
305	THETA_K_MORT_AKI	1.020	-	Arhenius temperature correction factor for akinete mortality rate constant
306	T_GERM_AKI	21.0	*deg C*	Temperature to trigger germination when DIN is also low

Literature References:

[1] Paerl, H.W. & Otten, T.G. (2013)

As above.

Relevance: Growth ecology of Nostocales; mu_max 0.5-2.0 /day.

[2] Hense, I. & Beckmann, A. (2006)

Towards a model of cyanobacteria life cycle -- effects of growing and resting stages on bloom formation of N2-fixing species. Ecological Modelling, 195, 205-218.

Relevance: Akinete germination rates 0.1-0.5 /day; formation triggered by T < 16 deg C and declining DIN. Model uses similar parameterisation.

[3] Sukenik, A., Hadas, O. & Kaplan, A. (2012)

Invasion of Nostocales (cyanobacteria) to subtropical and temperate freshwater lakes -- physiological, regional, and global driving forces. Frontiers in Microbiology, 3, 86.

Relevance: Nostocales temperature requirements T_opt 24-28 deg C; T_min 15-20 deg C.

[4] Kovacs, A.W., Presing, M. & Voros, L. (2016)

Thermal-dependent growth characteristics for Cylindrospermopsis raciborskii. Hydrobiologia, 764(1), 97-108.

Relevance: CTMI cardinal temperatures for Nostocales species.

Dissolution Saturation & DOM Availability

Half-saturation constants for POM dissolution saturation, available fractions of DON/DOP for Nostocales, and phytoplankton-dependent mineralisation caps.

Value Selection Rationale:

KHS_POC_DISS_SAT = 1.25 mg C/L prevents runaway dissolution at high POC. frac_avail_DON/DOP = 0.0: DON/DOP not directly bioavailable to phytoplankton (must be mineralised first). K_MIN_PHYT_AMIN caps = 4.0 limit phytoplankton-enhanced mineralisation (Wetzel, 2001).

#	Constant Name	Value	Unit	Description
307	`KHS_POC_DISS_SAT`	1.250	*mg C/L*	
308	`KHS_PON_DISS_SAT`	0.250	*mg N/L*	
309	`KHS_POP_DISS_SAT`	0.0250	*mg P/L*	
310	`frac_avail_DON`	0.00	-	
311	`frac_avail_DOP`	0.00	-	
312	`frac_avail_DON_NOST`	0.0	-	
313	`KHS_DN_NOST_VEG_HET`	0.00720	*mg N/L*	half saturation constant for limitation of nost. growth on nitrogen
314	`FRAC_FIX_N_FOR_GR_VEG_HET`	0.650	-	fraction of fixed ntrogen used for nostacles for growth. 1-frac is released to environment
315	`FRAC_NOST_GROWTH`	0.10	-	fraction of maximum growth rate supported by fixed nitrogen 1-frac - supported by N forms in WC
316	`K_MIN_PHYT_AMIN_DOC`	4.00	*L/(mg C)*	
317	`K_MIN_PHYT_AMIN_DON`	4.00	*L/(mg C)*	
318	`K_MIN_PHYT_AMIN_DOP`	4.00	*L/(mg C)*	

Literature References:

[1] Wetzel, R.G. (2001)

As above.

Relevance: Saturation kinetics for POM dissolution.

Photoinhibition (beta)

Photoinhibition parameters for each phytoplankton group. beta = 0 uses the default Steele (1962) formulation; beta > 0 increases photo-inhibition at high irradiance.

Value Selection Rationale:

All BETA values = 0.0: default Steele curve P = (I/I_s)*exp(1 - I/I_s) is used. This is appropriate when surface photoinhibition is negligible or the model is applied to well-mixed water columns (Platt et al., 1980).

#	Constant Name	Value	Unit	Description
319	`BETA_DIA`	0.0	-	Diatoms photoinhibition (0=Steele default, >0=stronger)
320	`BETA_CYN`	0.0	-	Non-fixing cyanobacteria photoinhibition
321	`BETA_FIX_CYN`	0.0	-	Fixing cyanobacteria photoinhibition
322	`BETA_OPA`	0.0	-	OtherPhyto photoinhibition
323	`BETA_NOST_VEG_HET`	0.0	-	Nostocales photoinhibition

Literature References:

[1] Steele, J.H. (1962)

Environmental control of photosynthesis in the sea. Limnology and Oceanography, 7(2), 137-150.

Relevance: Classic photoinhibition formulation P/P_max = (I/I_s)*exp(1 - I/I_s). AQUABC default beta = 0 uses this curve; beta > 0 intensifies inhibition.

[2] Platt, T., Gallegos, C.L. & Harrison, W.G. (1980)

Photoinhibition of photosynthesis in natural assemblages of marine phytoplankton. J. Marine Research, 38, 687-701.

Relevance: Extended P-I curve with explicit photoinhibition parameter beta.

Complete Bibliography

[1] Bastviken, D. et al. (2004) Methane emissions from lakes: dependence of lake characteristics, two regional assessments, and a global estimate. Global Biogeochemical Cycles, 18(4), GB4009.

[2] Bernard, O. & Rémond, B. (2012) Validation of a simple model accounting for light and temperature effect on microalgal growth. Bioresource Technology, 123, 520-527.

[3] Berner, R.A. (1980) Early Diagenesis: A Theoretical Approach. Princeton Univ. Press.

[4] Boudreau, B.P. (1997) Diagenetic Models and Their Implementation. Springer.

[5] Bowie, G.L. et al. (1985) As above -- EPA compilation.

[6] Bowie, G.L., Mills, W.B., Porcella, D.B. et al. (1985) Rates, Constants, and Kinetics Formulations in Surface Water Quality Modeling (2nd ed.). EPA/600/3-85/040. U.S. EPA, Athens, GA.

[7] Brzezinski, M.A. (1985) The Si:C:N ratio of marine diatoms: interspecific variability and the effect of some environmental variables. J. Phycology, 21(3), 347-357.

[8] Cerco, C.F. (2000) Phytoplankton kinetics in the Chesapeake Bay eutrophication model. Water Quality and Ecosystems Modeling, 1, 5-49.

[9] Chapra, S.C. (1997) Surface Water-Quality Modeling. McGraw-Hill, New York.

[10] Davison, W. (1993) Iron and manganese in lakes. Earth-Science Reviews, 34(2), 119-163.

[11] Eppley, R.W. (1972) Temperature and phytoplankton growth in the sea. Fishery Bulletin, 70(4), 1063-1085.

[12] Falkowski, P.G. & Raven, J.A. (2007) Aquatic Photosynthesis, 2nd ed. Princeton Univ. Press.

[13] Geider, R.J., MacIntyre, H.L., & Kana, T.M. (1997) Dynamic model of phytoplankton growth and acclimation. Marine Ecology Progress Series, 148, 187-200.

[14] Grimaud, G.M., Mairet, F. et al. (2017) Modeling the temperature effect on the specific growth rate of phytoplankton: a review. Reviews in Environmental Science and Bio/Technology, 16, 625-645.

[15] Hansen, P.J., Bjornsen, P.K. & Hansen, B.W. (1997) Zooplankton grazing and growth: scaling within the 2-2000 um body size range. Limnology and Oceanography, 42, 687-704.

[16] Hense, I. & Beckmann, A. (2006) Towards a model of cyanobacteria life cycle -- effects of growing and resting stages on bloom formation of N2-fixing species. Ecological Modelling, 195, 205-218.

[17] Horne, A.J. & Goldman, C.R. (1994) Limnology, 2nd ed. McGraw-Hill.

[18] Jorgensen, B.B. (1982) Mineralization of organic matter in the sea bed -- the role of sulphate reduction. Nature, 296, 643-645.

[19] Jorgensen, S.E. & Bendoricchio, G. (2001) Fundamentals of Ecological Modelling, 3rd ed. Elsevier.

[20] Jorgensen, S.E. (1976) A eutrophication model for a lake. Ecological Modelling, 2(2), 147-165.

[21] Jorgensen, S.E. et al. (1991) Handbook of Ecological Parameters and Ecotoxicology. Elsevier.

[22] Kamatani, A. (1982) Dissolution rates of silica from diatoms decomposing at various temperatures. Marine Biology, 68, 91-96.

[23] Kirk, J.T.O. (1994) Light and Photosynthesis in Aquatic Ecosystems, 2nd ed. Cambridge Univ. Press.

[24] Kovacs, A.W., Presing, M. & Voros, L. (2016) Thermal-dependent growth characteristics for Cylindrospermopsis raciborskii. Hydrobiologia, 764(1), 97-108.

[25] Paerl, H.W. & Otten, T.G. (2013) Harmful cyanobacterial blooms: causes, consequences, and controls. Microbial Ecology, 65(4), 995-1010.

[26] Painter, H.A. (1970) A review of literature on inorganic nitrogen metabolism in microorganisms. Water Research, 4(6), 393-450.

[27] Platt, T., Gallegos, C.L. & Harrison, W.G. (1980) Photoinhibition of photosynthesis in natural assemblages of marine phytoplankton. J. Marine Research, 38, 687-701.

[28] Redfield, A.C. (1958) The biological control of chemical factors in the environment. American Scientist, 46(3), 205-221.

[29] Reynolds, C.S. & Irish, A.E. (1997) Modelling phytoplankton dynamics in lakes and reservoirs: the problem of in-situ growth rates. Hydrobiologia, 349, 5-17.

[30] Reynolds, C.S. (2006) The Ecology of Phytoplankton. Cambridge Univ. Press.

[31] Robarts, R.D. & Zohary, T. (1987) Temperature effects on photosynthetic capacity, respiration, and growth rates of bloom-forming cyanobacteria. New Zealand J. Marine and Freshwater Research, 21(3), 391-399.

[32] Rosso, L. et al. (1993) As above -- CTMI model for temperature response.

[33] Rosso, L., Lobry, J.R., & Flandrois, J.P. (1993) An unexpected correlation between cardinal temperatures of microbial growth highlighted by a new model. J. Theoretical Biology, 162(4), 447-463.

[34] Soetaert, K. et al. (1996) As above.

[35] Soetaert, K., Herman, P.M.J. & Middelburg, J.J. (1996) A model of early diagenetic processes from the shelf to abyssal depths. Geochimica et Cosmochimica Acta, 60(6), 1019-1040.

[36] Sommer, U. & Sommer, F. (2006) Cladocerans versus copepods: the cause of contrasting top-down controls on freshwater and marine

phytoplankton. Oecologia, 147, 183-194.

[37] Sommer, U. (Ed.) (1989) Plankton Ecology: Succession in Plankton Communities. Springer.

[38] Staal, M., Meysman, F.J.R. & Stal, L.J. (2003) Temperature excludes N2-fixing heterocystous cyanobacteria in the tropical oceans. Nature, 425, 504-507.

[39] Steele, J.H. (1962) Environmental control of photosynthesis in the sea. Limnology and Oceanography, 7(2), 137-150.

[40] Stumm, W. & Morgan, J.J. (1996) Aquatic Chemistry: Chemical Equilibria and Rates in Natural Waters, 3rd ed. Wiley.

[41] Sukenik, A., Hadas, O. & Kaplan, A. (2012) Invasion of Nostocales (cyanobacteria) to subtropical and temperate freshwater lakes -- physiological, regional, and global driving forces. Frontiers in Microbiology, 3, 86.

[42] Van Cappellen, P. & Wang, Y. (1996) Cycling of iron and manganese in surface sediments: a general theory for the coupled transport and reaction of carbon, oxygen, nitrogen, sulfur, iron, and manganese. American Journal of Science, 296, 197-243.

[43] Wetzel, R.G. (2001) Limnology: Lake and River Ecosystems, 3rd ed. Academic Press.

Additional Key References for Water Quality Modelling:

[44] Ambrose, R.B., Wool, T.A. & Martin, J.L. (1993) The Water Quality Analysis Simulation Program, WASP5. Part A: Model Documentation. U.S. EPA, Athens, GA.

[45] Brown, L.C. & Barnwell, T.O. (1987) The Enhanced Stream Water Quality Models QUAL2E and QUAL2E-UNCAS. EPA/600/3-87/007.

[46] Cole, T.M. & Wells, S.A. (2006) CE-QUAL-W2: A Two-Dimensional, Laterally Averaged, Hydrodynamic and Water Quality Model, Version 3.5. U.S. Army Corps of Engineers.

[47] Collins, C.D. & Wlosinski, J.H. (1983) Coefficients for Use in the US Army Corps of Engineers Reservoir Model CE-QUAL-R1. Report E-83-15.

[48] Di Toro, D.M. (2001) Sediment Flux Modeling. Wiley-Interscience.

[49] Eppley, R.W. (1972) Temperature and phytoplankton growth in the sea. Fishery Bulletin, 70(4), 1063-1085.

[50] Hamilton, D.P. & Schladow, S.G. (1997) Prediction of water quality in lakes and reservoirs. Part I -- Model description. Ecological Modelling, 96, 91-110.

[51] James, R.T., Martin, J., Wool, T. & Wang, P.F. (1997) A sediment resuspension and water quality model of Lake Okeechobee. JAWRA, 33(3), 661-680.

[52] Jorgensen, S.E. & Bendoricchio, G. (2001) Fundamentals of Ecological Modelling, 3rd ed. Elsevier.

[53] Lindim, C., Pinho, J.L. & Vieira, J.M.P. (2011) Analysis of spatial and temporal patterns in a large reservoir using water quality and hydrodynamic modeling. Ecological Modelling, 222, 2485-2494.

[54] Sakurova, I. (2019) Modeling of eutrophication processes in the Curonian Lagoon. Master's Thesis, Klaipeda University. [ESTAS-AQUABC model application].

[55] Idzelyte, R. (2016) A visualisation system for biogeochemical and hydrodynamic modelling and its application to the Curonian lagoon. Master's Thesis, Klaipeda University.

[56] Eilola, K., Almroth-Rosell, E., Edman, M. et al. (2015) Model set-up at COCOA study sites. SMHI Report. [AQUABC ecological model description].